

Nouraldin Farge

SOFTWARE ENGINEER — DESKTOP & LOCAL-FIRST SYSTEMS

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SUMMARY

Software engineer who shipped three verified portable Windows releases and a source-free engineering case study using React/TypeScript, Rust/Tauri, Electron, Python, and SQLite. Owns architecture, safety boundaries, verification, and release decisions for local-first systems.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Rust, Python, SQL, HTML/CSS

Desktop & Frontend: React, Tauri, Electron, Windows, WebView2, Vite

Data & Integration: SQLite, Node.js, Express, REST, WebSockets, JSON Schema, Zod

Quality & Release: GitHub Actions, CodeQL, Vitest, pytest, Playwright, Cargo/Clippy, SBOMs, SHA-256, provenance attestations

SELECTED PROJECTS

DrawScope v0.6.5 | React · Rust/Tauri · Python · SQLite

[Code](#) · [Release](#) · 2026–Present

- Built a portable historical-research workbench around a reproducible 41,598-row SQLite archive spanning six lottery games, with row-level provenance and explicit data gaps.
- Prevented look-ahead bias across 250 walk-forward targets with a fixed 60/40 discovery–confirmation split and contract tests across TypeScript, Python, and Rust.

GameVault v0.3.5 | React · Rust/Tauri · SQLite

[Code](#) · [Release](#) · 2026–Present

- Developed a portable game library and launcher with relative-path SQLite state, launch history, backups, and optional metadata restricted to allowlisted official store URLs.
- Hardened review-gated ZIP intake against traversal, unsafe Windows paths, links/reparse entries, and excessive expansion; added explicit promotion approval and rollback.

Day-Trading Teacher v0.32.6 | React · Rust/Tauri · local JSON · Zod

[Code](#) · [Release](#) · 2026–Present

- Created a lesson-first environment connecting decision plans, historical replay, paper practice, journaling, reflection, and spaced retrieval in one local learning loop.
- Implemented deterministic Rust decimal calculations, schema-validated lesson imports, and provenance-aware Fidelity history review while excluding live signals, advice, and orders.

Research Studio alpha.21 | Electron · React · TypeScript · Express · SQLite

[Case study](#) · 2026–Present

- Architected a private Chinese/English catalog-enrichment alpha with read-only snapshots, explicit working copies, batches of at most five, untrusted-output validation, human approval, and recoverable exports; published only a source-free case study.

EXPERIENCE

Independent Software Engineer

2026–Present

Portfolio Products

- Owned requirements through release for three public products, including architecture, trust boundaries, automated checks, portable recovery, documentation, checksums, SBOMs, and provenance.
- Directed AI-assisted implementation while personally reviewing changes, resolving failures, validating claims, and making safety, licensing, data-source, and release decisions.

Technical Assistant (Additional Experience)

Jun 2015–Sep 2015

Chicago Public Schools

Chicago, IL

- Supported staff technology and classroom systems, diagnosed hardware and software issues, and communicated practical resolutions to non-technical users.

EDUCATION

Northeastern Illinois University

May 2023

B.A. Computer Science, Cum Laude — GPA: 3.74

Chicago, IL